Notes and Solutions

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January 6, 2021

0.1 GFKG: Maxwell's Equations

1. Let $\vec{k}$ be a vector in $\mathbb{R}^3$ and let $\omega = |\vec{k}|$. Fix $\vec{E} \in \mathbb{C}^3$ with $\vec{k} \cdot \vec{E} = 0$ and $i\vec{k} \times \vec{E} = \omega \vec{E}$. Show that

$$\vec{\mathcal{E}}(t, \vec{x}) = \vec{E} e^{-i(\omega t - \vec{k} \cdot \vec{x})}$$

satisfies the vacuum Maxwell equations.

Proof (vacuum Maxwell plane wave). Let $\vec{k} = \begin{pmatrix} k^x \\ k^y \\ k^z \end{pmatrix}$ and $\vec{E} = \begin{pmatrix} E^x \\ E^y \\ E^z \end{pmatrix}$.

We compute the partial derivative with respect to time below:

$$\frac{\partial \vec{\mathcal{E}}}{\partial t} = \vec{E} e^{-i(-\vec{k} \cdot \vec{x})} \frac{\partial}{\partial t} \left(e^{-i(\omega t)} \right) = \vec{E} e^{-i(-\vec{k} \cdot \vec{x})} \left(-i\omega e^{(-i\omega)t} \right) = -i\omega \vec{\mathcal{E}}.$$

and of a spatial component $\mu = x, y,$ or z with respect to a spatial component z below:

$$\begin{aligned} \frac{\partial \mathcal{E}^\mu}{\partial z} &= E^\mu e^{-i(\omega t - k^x x - k^y y)} \frac{\partial}{\partial z} \left(e^{-i(-k^z z)} \right) = E^\mu e^{-i(\omega t - k^x x - k^y y)} \left(ik^z e^{(ik^z)z} \right) \\ &= ik^z E^\mu e^{-i(\omega t - \vec{k} \cdot \vec{x})}. \end{aligned}$$

The divergence is then given by:

$$\begin{aligned} \nabla \cdot \vec{\mathcal{E}} &= \frac{\partial \mathcal{E}^x}{\partial x} + \frac{\partial \mathcal{E}^y}{\partial y} + \frac{\partial \mathcal{E}^z}{\partial z} = ik^x E^x e^{-i(\omega t - \vec{k} \cdot \vec{x})} + ik^y E^y e^{-i(\omega t - \vec{k} \cdot \vec{x})} + ik^z E^z e^{-i(\omega t - \vec{k} \cdot \vec{x})} \\ &= i \left(k^x E^x + k^y E^y + k^z E^z \right) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = i \left(\vec{k} \cdot \vec{E} \right) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = i \left(0 \right) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = 0, \end{aligned}$$

and the curl is given by:

$$\begin{aligned} \nabla \times \vec{\mathcal{E}} &= \begin{pmatrix} \frac{\partial \mathcal{E}^z}{\partial y} - \frac{\partial \mathcal{E}^y}{\partial z} \\ \frac{\partial \mathcal{E}^x}{\partial z} - \frac{\partial \mathcal{E}^z}{\partial x} \\ \frac{\partial \mathcal{E}^y}{\partial x} - \frac{\partial \mathcal{E}^x}{\partial y} \end{pmatrix} = i \begin{pmatrix} k^y E^z - k^z E^y \\ k^z E^x - k^x E^z \\ k^x E^y - k^y E^x \end{pmatrix} e^{-i(\omega t - \vec{k} \cdot \vec{x})} \\ &= i(\vec{k} \times \vec{E}) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = (\omega \vec{E}) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = i \left(-i\omega \vec{E} e^{-i(\omega t - \vec{k} \cdot \vec{x})} \right) = i \frac{\partial \vec{\mathcal{E}}}{\partial t}. \end{aligned}$$

Therefore, $\vec{\mathcal{E}}$ satisfies the vacuum Maxwell equations; $\nabla \cdot \vec{\mathcal{E}} = 0$ and $\nabla \times \vec{\mathcal{E}} = i \frac{\partial \vec{\mathcal{E}}}{\partial t}$. □